

Olist E-Commerce Marketplace Analysis

What drives revenue — and what threatens scaling. A deep-dive into 100K orders across Brazil, 2016–2018.

BRAZIL · 100K ORDERS · 2016–2018

SQL · PYTHON · POWER BI

The Marketplace Is Growing. But Is This Growth Healthy?

13.2M

BRL Revenue

Total across all delivered orders

96,478

Delivered Orders

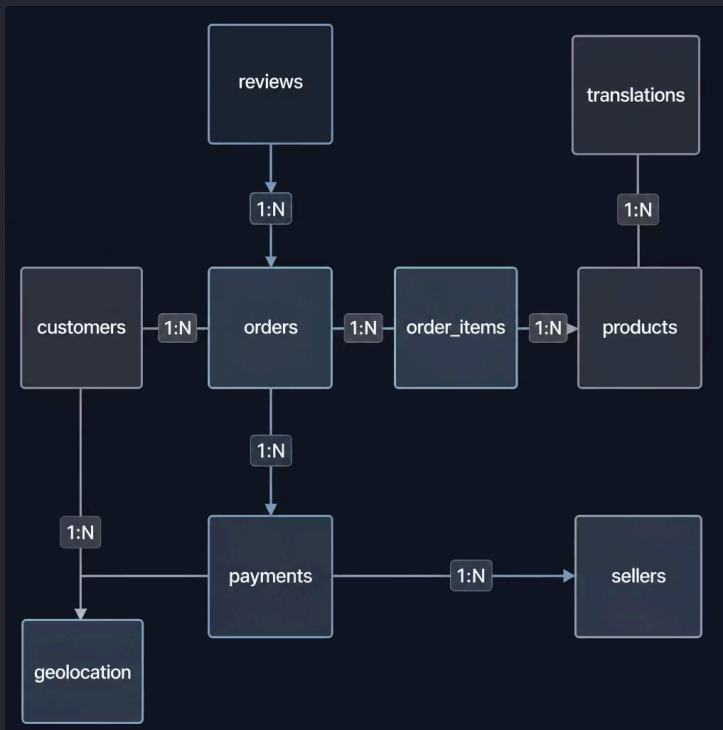
97% of total order volume

2,970

Active Sellers

Across all product categories

Three critical questions frame this analysis: Where does revenue really come from — and how concentrated is it? Do customers return, or must every sale be acquired from scratch? What breaks the customer experience, and who owns the problem?



From 9 Raw CSV Files to an Executive Dashboard

The Data Pipeline

Nine related CSV tables (~100K orders) were modeled into a relational schema (ERD), then processed through a structured pipeline:

❏ Scope was deliberate: all KPIs computed on **delivered orders only** — 97% of volume, excluding cancelled and undelivered orders.

- ➔ **Clean** — pandas / Power Query preprocessing
- ➔ **Model** — SQLite with window functions and CTEs
- ➔ **Validate** — 12 cross-validated KPIs
- ➔ **Visualize** — 3-page Power BI dashboard with DAX measures

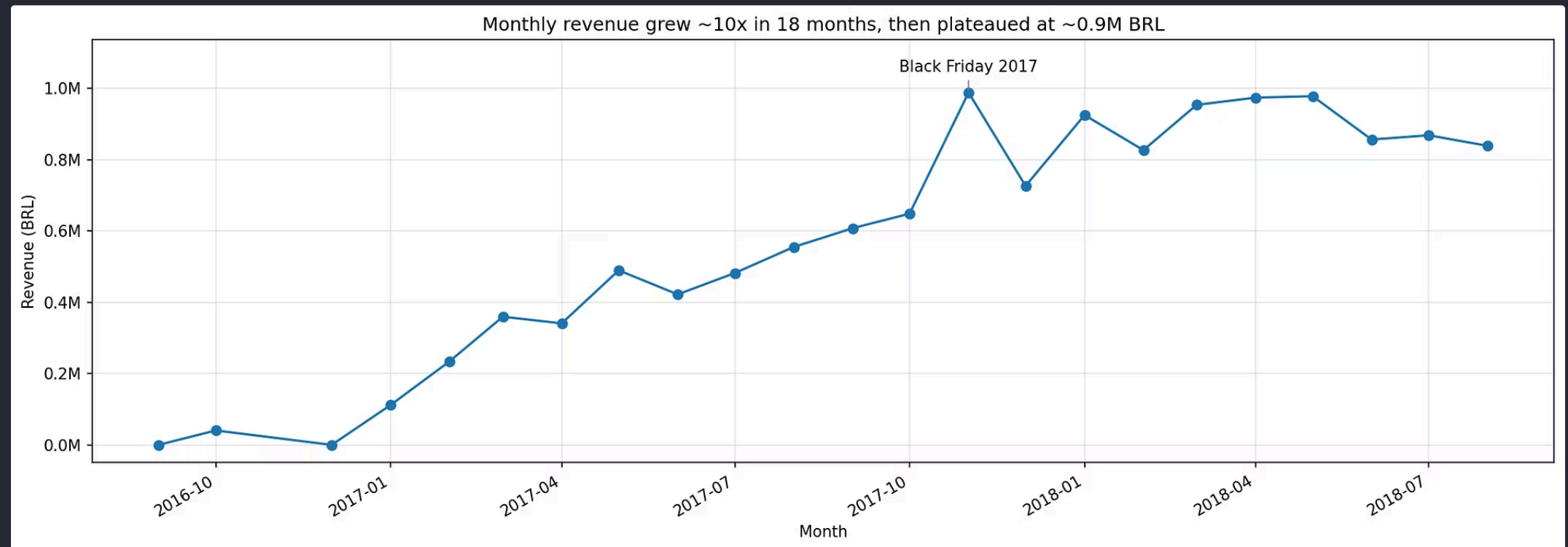
Sales Run on New Customers

Retention Is Critically Low

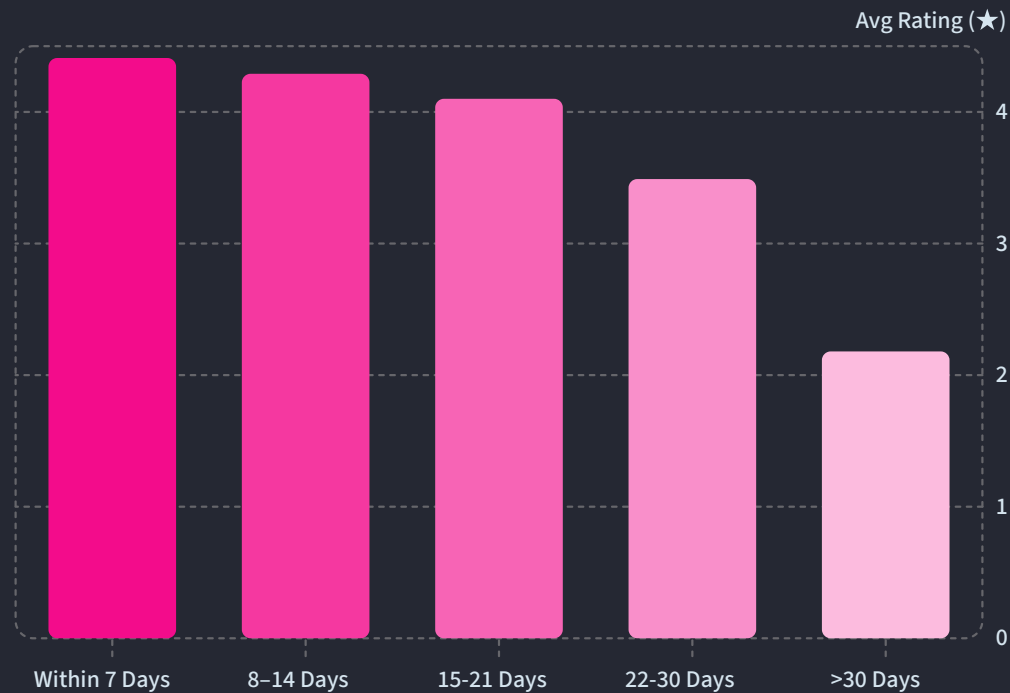
Only **3.0%** of customers ever place a second order — averaging just **1.03 orders per customer** across the entire dataset.

Black Friday 2017 delivered record sales, yet only **1.9%** of that cohort returned.

96.8% of orders are first purchases. Whether this growth is expensive depends on acquisition costs and margins, which the data does not include.



Customer Satisfaction Is Set by Logistics, Not the Product



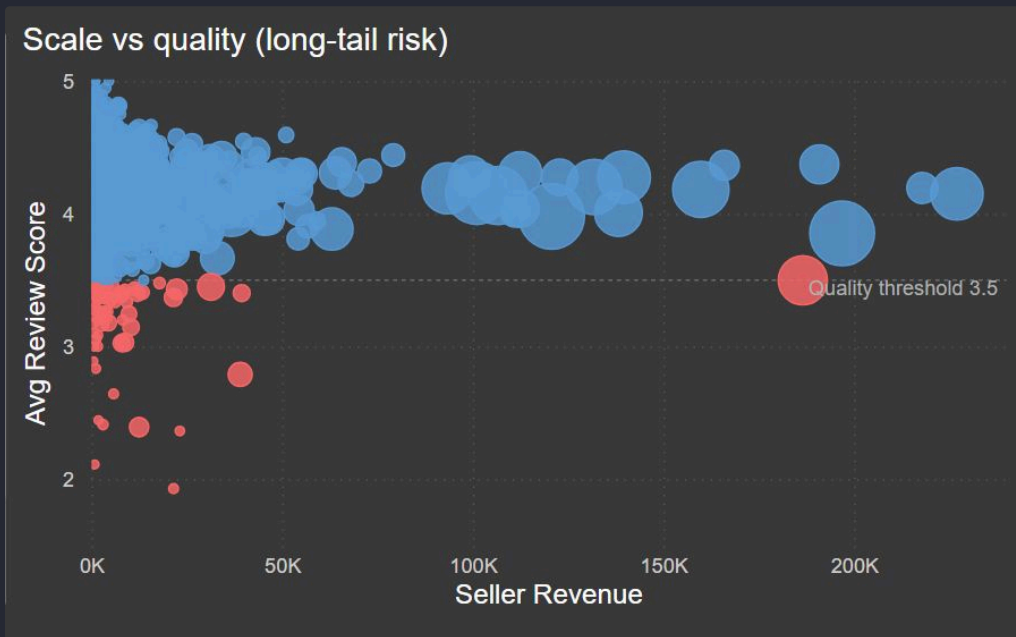
Delivery Speed Drives Ratings

Customers who receive orders within 7 days rate 4.41★ on average. Ratings hold above 4.1★ up to 21 days — then satisfaction collapses: 3.49★ at 22–30 days, and just 2.18★ beyond 30 days.

The root cause is geographic: São Paulo averages **8.3 days** delivery, while northern states take **26–27 days** — nearly 3× slower.

Logistics, not product quality, is the primary lever for customer satisfaction improvement.

Big Sellers Make the Revenue. Small Sellers Make the Risk.



Concentration Confirmed

Across 2,970 active sellers, the classic 80/20 rule holds: the **top 20%** generate **82.3%** of all revenue.

Yet **343 sellers** (1 in 9) carry an average rating below 3.5★ — almost exclusively in the long tail. Most large sellers rate above 4.0★, but scale does not guarantee quality: 3 sellers above 100K BRL revenue average below 4.0★, the lowest at 3.5★ with 187K BRL.

Extreme tail case: **1.93★ average rating** with **64% late deliveries**.

Three Levers to Unlock Sustainable Growth



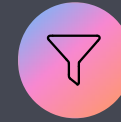
Retain the Customer

Post-purchase loyalty program: a coupon for a complementary category, valid 30 days after delivery — conditional on a positive review.



Deliver on Time

SLA with a hard **21-day limit** (the pain threshold found in data) and a **14-day operational target** for fulfillment teams.



Filter Weak Sellers

Quality gate at onboarding + automatic alert when a seller's rating drops below 3.5★ — before they damage platform trust.

What This Project Taught Me

Cross-Validation Catches Real Bugs

Power BI flagged 348 "problem sellers"; SQL returned 343. The gap of 5 traced to a DAX bug: `BLANK < 3.5` evaluates TRUE, falsely flagging sellers with no reviews.

Scope Is a Decision, Not an Assumption

Every KPI was deliberately computed on delivered orders only. The same metric with a different filter is a different number — both can be "correct."

Next Steps

Cost data (CAC, margins) would convert directional recommendations into a business case with ROI — followed by pipeline automation.

github.com/wp-pakulski/olist-ecommerce-analysis